

All-degree angle transport for the original arithmetic four-volume

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The original finite quotient has two pairs of minimum-section subspaces. This calculation identifies their complete principal angles with the energies of their actual relation-valued differences. Their determinant is the original arithmetic four-volume. The original vertical-line moment identity then proves strict positivity of that four-volume and gives a nonlinear endpoint bound in every quotient dimension.

1 Fixed arithmetic source, maps and forms

Fix the original full actual packet h for $g = 2\xi$, an integer $k \geq 1$, and its original cyclic polynomial $\chi = \chi_{h,k}$, monic of degree $q \geq 1$. Keep $E = \mathbb{C}[S]/(\chi)$ in its increasing-power remainder frame. The full Taylor unit $v_h = j_h(g/h)$ occurs in the unchanged injections

$$\eta : E \hookrightarrow E_h^{\otimes k}, \quad \eta[P] = v_h^{\otimes k} P(A_h^{(k)})1, \quad \sigma_h^{\otimes k} \eta : E \hookrightarrow Q^{\otimes k}, \quad (\text{AGT1})$$

where $E_h = \mathbb{C}[s]/(h)$ and $Q = \mathcal{B}/\Theta V$. The latter is the top cohomology of the original tensor pushforward over $\mathbf{F}_{1,\tau}$. All these maps and the original theta primitives are supplied, with their complete proofs, in the retained AT and AW dependency packages.

The common polynomial source is $H = \mathcal{P}_{2q}$, with coordinate $S = k/2 + iu$ on the original integration line. Its original measures are

$$\begin{aligned} m_1(u) &= w_h^{*k}(u), & w_h(u) &= \frac{|(g/h)(1/2 + iu)|^2}{2\pi}, \\ m_0(u) &= \frac{(\sqrt{2\pi})^k}{c_{k/4}} \frac{|\Gamma(k/4 + iu/2)|^2}{2\pi}, & c_a &= 2^{1-2a}\Gamma(2a). \end{aligned} \quad (\text{AGT2})$$

Their full masses $(2\pi)^{k/2}$ and μ_h^k remain present. Set $m_t = (1-t)m_0 + tm_1$, $0 \leq t \leq 1$. In the original coefficient frame write $M(t)$ for its Gram on H , and $\langle x, y \rangle_t = x^* M(t) y$. The complete source-density proofs give all finite moments and positivity almost everywhere. Therefore this form is positive on every nonzero polynomial, for every t in the closed interval.

For each $q-1 \leq N \leq 2q$, use the original exact sequence $0 \rightarrow \mathcal{P}_{N-q} \xrightarrow{\times \chi} \mathcal{P}_N \xrightarrow{J_N} E \rightarrow 0$, with $\mathcal{P}_{-1} = 0$. Let $R_N(t) : E \hookrightarrow \mathcal{P}_N \subset H$ be its unique minimum-norm section, and retain

$$G_N(t) = R_N(t)^* M(t) R_N(t), \quad V_N(t) = \det G_N(t). \quad (\text{AGT3})$$

The inclusions into H are part of this formula. Write P_N, Q_N for the orthogonal projections in H onto $\mathcal{P}_N$ and $\chi \mathcal{P}_{N-q}$ respectively, and $Z_N = P_N - Q_N$. Then Z_N projects onto $R_N E$ and has rank q .

The existing complete determinant-variation proof gives

$$\begin{aligned} \mathcal{B}(t) &= \log \frac{V_{q-1}(t) V_q(t)}{V_{2q-1}(t) V_{2q}(t)}, \\ C(t) &= M(t)^{-1} (M(1) - M(0)), \\ \dot{\mathcal{B}}(t) &= \text{Tr}((Z_{q-1} - Z_{2q} + Z_q - Z_{2q-1}) C(t)). \end{aligned} \quad (\text{AGT4})$$

This groups the same four signed terms into two pairs; all four original cutoff indices and quotient Grams remain fixed.

2 Relation energy, Gram ratios and the full principal angles

Theorem 1. For $q - 1 \leq i \leq j \leq 2q$, define the actual difference of minimum sections $D_{ij} = R_i - R_j : E \rightarrow H$. Then

$$\begin{aligned} D_{ij}E &\subset \chi\mathcal{P}_{j-q}, & R_j &= Z_j R_i, \\ G_i - G_j &= D_{ij}^* M D_{ij}, & K_{ij} &= G_i^{-1/2} G_j G_i^{-1/2}, \\ I - K_{ij} &= G_i^{-1/2} D_{ij}^* M D_{ij} G_i^{-1/2}, & 0 &\prec K_{ij} \preceq I. \end{aligned} \quad (\text{AGT5})$$

The eigenvalues of K_{ij} , including multiplicities, are $\cos^2 \theta_{ij,a}$ for the full principal-angle list between $R_i E$ and $R_j E$, with $0 \leq \theta_{ij,a} < \pi/2$ and $1 \leq a \leq q$. Consequently

$$\log(V_i/V_j) = \sum_{a=1}^q -\log \cos^2 \theta_{ij,a}. \quad (\text{AGT6})$$

The nonzero eigenvalues of $Z_i - Z_j$ are exactly the pairs $\sin \theta_{ij,a}, -\sin \theta_{ij,a}$ for the nonzero angles; the remaining eigenvalues are zero.

Proof. Both sections have the same remainder, so their difference lies in the displayed relation space. The vector $R_j x$ is orthogonal to that entire space. Also $R_i x \in \mathcal{P}_j$. Thus the orthogonal splitting of $R_i x$ into its relation and minimum-lift components is $D_{ij} x + R_j x$, proving $R_j = Z_j R_i$ and the Gram identity. Positivity of G_j and this nonnegative difference prove the last inequalities in (AGT5). Each square root is the unique positive Hermitian square root on the displayed original coefficient space; the order of its products is retained.

For the principal-angle calculation put $U_i = R_i G_i^{-1/2}$ and $U_j = R_j G_j^{-1/2}$. They are isometries from the displayed Euclidean coefficient space to the two original source subspaces: $U_i^* M U_i = I$ and $U_j^* M U_j = I$. Their full cross matrix is

$$U_j^* M U_i = G_j^{-1/2} R_j^* M R_i G_i^{-1/2} = G_j^{1/2} G_i^{-1/2}. \quad (\text{AGT7})$$

The equality uses $R_j^* M D_{ij} = 0$. Its squared singular values are therefore the eigenvalues of K_{ij} , with no commutation of the two Grams presumed. These singular values are positive and at most one, defining the angles just stated. Determinants of the displayed positive matrices give (AGT6), keeping all original determinant factors.

Here is also a direct proof of the projection-difference spectrum. Choose singular-vector bases e_a in $R_i E$ and f_a in $R_j E$ for (AGT7), so their cross inner products are $\langle f_a, e_b \rangle_t = c_a \delta_{ab}$, with $c_a = \cos \theta_{ij,a} > 0$. For $c_a = 1$, the equality of norms gives $e_a = f_a$. For $c_a < 1$, put $s_a = \sqrt{1 - c_a^2}$ and $r_a = (f_a - c_a e_a)/s_a$. Substitution in these inner products proves that all resulting pairs (e_a, r_a) are orthonormal, orthogonal to the other pairs and to the common vectors. On each pair the exact matrix is

$$(Z_i - Z_j)|_{\text{span}(e_a, r_a)} = \begin{pmatrix} s_a^2 & -s_a c_a \\ -s_a c_a & -s_a^2 \end{pmatrix}. \quad (\text{AGT8})$$

It has trace zero and determinant $-s_a^2$, hence eigenvalues $s_a, -s_a$. The difference vanishes on the common vectors and on the orthogonal complement of the two subspaces' sum. This proves the full spectrum, including all zero-angle cases. $\square$

The difference D_{ij} in (AGT5) is an original polynomial relation. The original source map $\mathcal{V}$ therefore gives its literal theta primitive, while its full-jet and arithmetic cohomology observations are zero. Precisely, $J_j D_{ij} = 0$, $\eta J_j D_{ij} = 0$ and $\sigma_h^{\otimes k} \eta J_j D_{ij} = 0$; nevertheless its source energy is exactly $G_i - G_j$ in (AGT5). At each nonempty original support label A , the corresponding linear maps lift as $(A, x) \mapsto (A, fx)$, with $\tau \mapsto \tau$. The boundary then has a supported-zero class at that same label, and (AGT5) retains its entire preimage energy.

3 The actual four-volume has at most $2q - 1$ nonzero angles

Apply the theorem to the two pairs $(q - 1, 2q)$ and $(q, 2q - 1)$. For the second pair both q -dimensional minimum-section subspaces lie in

$$\mathcal{P}_{2q-1} \cap (\chi\mathcal{P}_0)^\perp, \quad \dim(\mathcal{P}_{2q-1} \cap (\chi\mathcal{P}_0)^\perp) = 2q - 1. \quad (\text{AGT9})$$

Indeed both are orthogonal to $\chi\mathcal{P}_0$, a one-dimensional subspace of $\mathcal{P}_{2q-1}$. Their intersection thus has dimension at least one, by the dimension formula for a sum of subspaces. Each common unit vector gives singular value one in the principal-angle calculation, so this pair has at least one exact zero angle. When $q = 1$ its two cutoff indices coincide, and its single angle is exactly zero.

Retain both full lists of q angles. Their combined logarithmic cost from (AGT6) is exactly $\mathcal{B}(t)$. Remove one of the proved zero contributions only in the following scalar list, and write the remaining $m = 2q - 1$ entries as

$$x_\ell(t) = -\log \cos^2 \theta_\ell(t) \geq 0, \quad \sum_{\ell=1}^m x_\ell(t) = \mathcal{B}(t), \quad \sin \theta_\ell(t) = \sqrt{1 - e^{-x_\ell(t)}}. \quad (\text{AGT10})$$

Other zero angles remain in the list. The zero-angle eigenspaces and the full source maps were not removed from their original spaces. No differentiability of a chosen eigenvector or ordered angle branch will be needed.

Let $c_{\min}(t), c_{\max}(t)$ be the extreme eigenvalues of $C(t)$ for $M(t)$. A self-adjoint projection difference with the spectrum in (AGT8) is a sum of its positive and negative rank-one eigenprojections with coefficients $s_a, -s_a$. Each eigenprojection's trace against $C(t)$ is the Rayleigh quotient of $C(t)$ at its unit vector and lies between these two extremes. Subtracting each pair gives an absolute value at most $s_a(c_{\max} - c_{\min})$. Apply this to both terms of (AGT4), retaining their signs, to obtain

$$|\dot{\mathcal{B}}(t)| \leq \left(\sum_{\ell=1}^m \sqrt{1 - e^{-x_\ell(t)}} \right) (c_{\max}(t) - c_{\min}(t)). \quad (\text{AGT11})$$

This first estimate retains all principal angles through the actual source relation energies in (AGT5).

4 The moment identity forces strict positivity

Write the original polynomial as $\chi(S) = \sum_{j=0}^q a_j S^j$, $a_q = 1$. The antilinear algebra involution on $\mathbb{C}[S]$ defined by

$$\chi^{\#k}(S) = \sum_{j=0}^q \bar{a}_j (k - S)^j \quad (\text{AGT12})$$

satisfies $\chi^{\#k}(S) = \overline{\chi(\bar{S})}$ on the original line, because $\bar{S} = k - S$. Its degree is q and its leading coefficient is $(-1)^q$. Applying it twice restores each coefficient and the variable since k is real.

Each term in (AGT10) is nonnegative, so $\mathcal{B}(t) \geq 0$. If it were zero, all angles of the first pair would be zero, hence $R_{q-1}E = R_{2q}E$. The first space is $\mathcal{P}_{q-1}$; the second is $(\chi\mathcal{P}_q)^\perp$ in H . It would follow that 1 is orthogonal to $\chi\mathcal{P}_q$. But $\chi^{\#k} \in \mathcal{P}_q$, and therefore this orthogonality would imply

$$0 = \langle 1, \chi\chi^{\#k} \rangle_t = \int_{\mathbb{R}} |\chi(k/2 + iu)|^2 m_t(u) du > 0. \quad (\text{AGT13})$$

The last strict inequality is positivity of the original form on the nonzero polynomial χ . This contradiction proves $\mathcal{B}(t) > 0$ at every $t \in [0, 1]$. Thus the strict positivity follows from an exact identity in the actual moment form; an arbitrary positive coefficient form has not been substituted for that form.

5 A nonlinear endpoint estimate in every degree

Theorem 2. Put $m = 2q - 1$, let $b_{\min}, b_{\max}$ be the extreme eigenvalues of the original relative Gram $M(0)^{-1}M(1)$, and set $\kappa = b_{\max}/b_{\min}$. Then the original two endpoint four-volumes satisfy

$$\left| 2 \operatorname{arcosh}\left(e^{\mathcal{B}(1)/(2m)}\right) - 2 \operatorname{arcosh}\left(e^{\mathcal{B}(0)/(2m)}\right) \right| \leq \log \kappa. \quad (\text{AGT14})$$

In particular this holds for every actual full quartet quotient, with its unchanged value of q .

Proof. The function $f(x) = \sqrt{1 - e^{-x}}$ is continuous for $x \geq 0$, with $f(0) = 0$. For $x > 0$ its two derivatives give

$$f'(x) = \frac{e^{-x}}{2\sqrt{1 - e^{-x}}}, \quad f''(x) = -\frac{e^{-x}}{2\sqrt{1 - e^{-x}}} - \frac{e^{-2x}}{4(1 - e^{-x})^{3/2}} < 0. \quad (\text{AGT15})$$

It is consequently concave on the whole closed half-line, by continuity at zero. More explicitly, its decreasing derivative shows that its tangent at any positive a is above its graph on $(0, \infty)$, and continuity extends this inequality to zero. Choose $a = \mathcal{B}(t)/m > 0$, proved admissible in (AGT13), and sum those tangent inequalities at the entries in (AGT10). Their linear terms cancel exactly, giving

$$\sum_{\ell=1}^m f(x_{\ell}(t)) \leq m\sqrt{1 - e^{-\mathcal{B}(t)/m}}, \quad |\dot{\mathcal{B}}(t)| \leq m\sqrt{1 - e^{-\mathcal{B}(t)/m}}(c_{\max}(t) - c_{\min}(t)). \quad (\text{AGT16})$$

This argument allows every remaining zero angle without differentiating it.

The positive finite Grams and their sections are smooth through the whole compact interpolation interval. Therefore $\mathcal{B}$ is smooth and, by (AGT13), has a positive minimum there. The real function $\Phi_m(B) = 2 \operatorname{arcosh}(e^{B/(2m)})$, using the nonnegative real inverse of the hyperbolic cosine, satisfies

$$\Phi'_m(B) = \frac{1}{m\sqrt{1 - e^{-B/m}}}, \quad B > 0. \quad (\text{AGT17})$$

One obtains this by differentiating $\cosh(\Phi_m(B)/2) = e^{B/(2m)}$. Multiply (AGT16) by this positive derivative.

To integrate the remaining spectral width, set $D = M(0)^{-1}M(1)$. It is positive and self-adjoint for the original $M(0)$ metric, and $C(t) = ((1-t)I + tD)^{-1}(D - I)$. Each relative eigenvalue $b > 0$ gives $c_b(t) = (b-1)/(1-t+tb)$. Its derivative with respect to b is $(1-t+tb)^{-2} > 0$, so the same extreme eigenvalues give the extremes for all t . Direct integration of the logarithmic derivative yields

$$\int_0^1 (c_{\max}(t) - c_{\min}(t)) dt = \int_0^1 \frac{d}{dt} \log \frac{1-t+tb_{\max}}{1-t+tb_{\min}} dt = \log \kappa. \quad (\text{AGT18})$$

Integrating $|(\Phi_m \circ \mathcal{B})'| \leq c_{\max} - c_{\min}$ proves (AGT14), including repeated relative eigenvalues. $\square$

An explicit interval for the original arithmetic endpoint is obtained without an implicit inversion. Put $A_0 = \operatorname{arcosh}(e^{\mathcal{B}(0)/(2m)})$ and $L = \frac{1}{2} \log \kappa$. Since $2m \log \cosh A$ is increasing on $A \geq 0$, (AGT14) is equivalent to

$$2m \log \cosh(\max\{0, A_0 - L\}) \leq \mathcal{B}(1) \leq 2m \log \cosh(A_0 + L). \quad (\text{AGT19})$$

The lower endpoint can be zero although the actual $\mathcal{B}(1)$ was proved strictly positive. Also $\Phi'_m(B) > 1/m$ for finite positive B , so (AGT14) implies $|\mathcal{B}(1) - \mathcal{B}(0)| \leq m \log \kappa$, with strict inequality in this last comparison whenever the two four-volumes differ. The full-relative-spectrum bound in AW13–AW14 remains available as a second, separately proved bound; none of its eigenvalues is discarded by the exact identities (AGT5)–(AGT11).

When $q = 1$, $m = 1$ and the second pair is identical. Equations (AGT14)–(AGT19) recover AW19–AW23 with exactly the original constant and mass factors. If the endpoint forms differ by a positive scalar only, then $\kappa = 1$; every quotient determinant has the same literal scalar to the power q , and the four-volume is unchanged, as the new inequality also proves.

6 Complete source and mathematical scope

The full AT and AW source packages accompany this calculation. Their complete proofs establish the original tau-base observation, tensor units, theta primitives, source densities, minimum sections and signed derivative (AGT4). The new principal-angle and moment calculations above retain those objects and prove (AGT5)–(AGT19) in every quotient dimension. The actual endpoint measures enter both the relative Gram D and the strict positivity argument. No upper estimate uniform in k for that relative Gram is assumed or claimed. The new bound supplies an explicit nonlinear constraint on the original arithmetic four-volume, with the complete source relation energies still present.